THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An iron nail kept in blue solution X gets reddish-brown coating and the solution turns green. X is likely: 1

(a) copper sulphate **(b)** sugar **(c)** sodium chloride **(d)** lime water

2. In a neutralisation, if 15 mL of acid is needed for 25 mL of a base, then for 50 mL of the base it will be: 1

(a) 60 mL **(b)** 30 mL **(c)** 25 mL **(d)** 15 mL

3. A tester lights when connected across tap water but only dimly across distilled water. Reason: 1

(a) tap water is pure **(b)** tap water has dissolved salts

(c) distilled water is wet **(d)** distilled water has salts

4. In $\mathrm{Fe} + \mathrm{CuSO_4} \longrightarrow \mathrm{FeSO_4} + \mathrm{Cu}$, the metal that displaces copper is: 1

(a) $\mathrm{FeSO_4}$ **(b)** Fe **(c)** $\mathrm{CuSO_4}$ **(d)** Cu

5. Which of the following would you use to show that a new gas is formed? 1

(a) lime water test **(b)** magnet **(c)** balance **(d)** ruler

6. Which statement about milk of magnesia is correct? 1

(a) it is a base used to relieve acidity **(b)** it is an acid used to relieve acidity

(c) it is a neutral salt used as a drink **(d)** it is an indicator

7. In the same tester the bulb does not glow when a dry wooden pencil case is used. The case is: 1

(a) a battery **(b)** an insulator **(c)** a conductor **(d)** a filament

8. Which of these metals is most suitable for making a very hot cooking pan? 1

(a) copper **(b)** sulphur **(c)** lead **(d)** wood

9. A farmer wants to keep tools from rusting during the rainy season. Which is best? 1

(a) wash daily with salt water **(b)** leave outside

(c) keep in a damp shed **(d)** apply oil and keep dry

10. Liquid A turns China rose pink, liquid B green. If A and B are mixed in equal strengths, the mixture most probably: 1

(a) becomes more acidic **(b)** turns almost neutral

(c) turns blue **(d)** becomes more basic

11. A chain of 10 decorative bulbs goes off when one bulb burns out. The bulbs are connected in: 1

(a) series **(b)** parallel **(c)** open circuit **(d)** no circuit

12. A metal Z reacts with dilute HCl giving hydrogen but does not react with cold water. Z may be: 1

(a) gold **(b)** sodium **(c)** zinc **(d)** potassium

13. **Assertion (A):** Melting of wax is a physical change. 1

Reason (R): No new substance is formed when wax melts.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** Red litmus paper turns blue when dipped in lemon juice. 1

Reason (R): Lemon juice is acidic in nature.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** A fuse is connected in series with an electrical appliance. 1

Reason (R): Fuse wire has a very high melting point, so it never melts.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are the pins of a plug made of brass or copper? 2

17. Why do non-metallic oxides cause acid rain? 2

18. Explain why the burning of paper is a chemical change but its tearing is a physical change. 2

19. Explain why a neutral solution does not change the colour of litmus. 2

20. Why do electric sockets have covers for children's safety? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why a mixture of salt and sand is physically separated but a mixture of iron and sulphur on heating is not. 3

22. Write the neutralisation equation for the formation of potassium sulphate. 3

23. Describe the safe way of fixing a bulb in a holder. 3

24. Why can't we make hammers from sulphur or wax? 3

OR

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Why is a firework display a mix of changes? 3

25. A soil test shows it is acidic. List three ways to treat it and explain. 3

OR

Compare the heating of a thin and a thick wire carrying the same current and explain why a thin fuse wire is used. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ananya's torch has two cells of 1.5 V each placed one after another, a small bulb and a push switch. One day, even after pressing the switch, the bulb does not glow, though the cells are new. 5

(i) What is the total voltage of the two cells joined in series?

(ii) Name the component that opens or closes the circuit.

(iii) Name the thin wire inside the bulb that glows.

(iv) Give one possible reason for the bulb not glowing and explain it.

27. Explain how a student can show that metal oxides are basic and non-metal oxides acidic using an indicator. 5

OR

Explain how to conduct an experiment to show that burning needs oxygen and produces carbon dioxide. 5

END OF QUESTION PAPER Total: 50 Marks

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